

DEPT OF ELECTRICAL & ELECTRONICS ENGINEERING

Program: T. Y. B. Tech ECE AI-ML Semester -6

Subject: Image Processing & Pattern Recognition (PC) (ECE3022B)

Experiment No.6

AIM: To perform pattern recognition using CNN or similar ML approach

Introduction

One of the techniques to recognize patterns is CNN: Convolution Neural Network. Hubel & Wiesel studied the structure of visual cortex in Mammals (won Nobel prize in 1981). Their research revealed that neurons in the visual cortex had a small local receptive field. i.e neurons look at a smaller subsection of the image that human is viewing. This idea then inspired an ANN architecture that is CNN: Convolution Neural Network. It was implemented in 1998 by Yann Le (Director, AI research at Facebook).

CNN mainly consists of two parts. First is feature extraction module which makes use of multiple Convolutional and pooling layers. Second module consists of fully connected layers which are trained for classification task. Figure 8.1 represents general architecture of CNN.

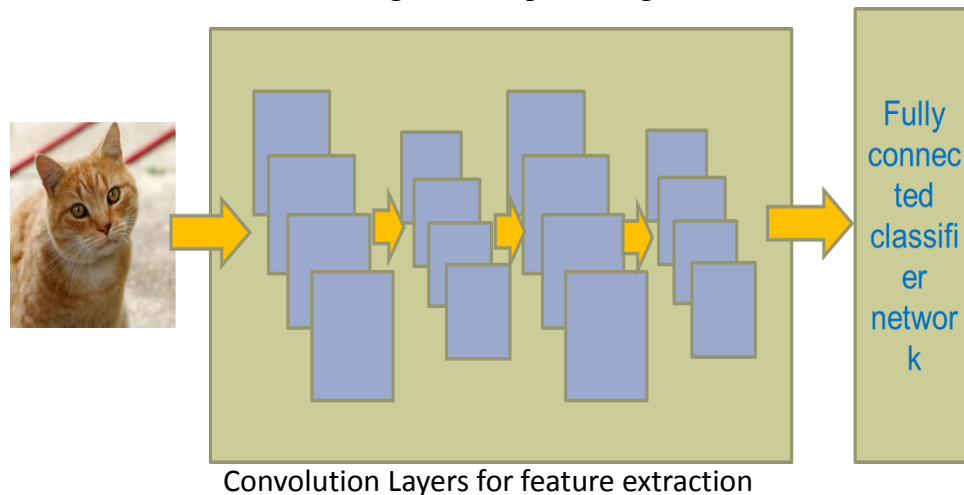


Fig. 8.1: Architecture of CNN

Algorithm:

1. Load data from MNIST dataset which includes images of handwritten digits 0 to 9 of size 28 x 28. [import mnist from keras.datasets]
2. Split the data into train and test datasets.
3. Pre-process the data to make sure the labels will be understandable by our CNN. MNIST dataset gives Labels as decimal values. For CNN we need to convert it to “one Hot Encoded” labels. [Use to_categorical function from keras.utils.np_utils]

4. Normalize the input data by dividing every input sample value by maximum value [255].
5. Reshape train and test samples to include one more dimension to represent the number of channels. As MNIST data represents grey images number of channels is one.
6. Set the CNN Model by selecting sequential model with Dense, Conv2D, MaxPool2D, Flatten layers.
7. Add 2D convolution layer with 32 number of filters each of size 4x4, input image shape 28 x 28 x 1 and activation function 'Relu'. [Use model.add]
8. Similarly add MaxPool layer with size 2 x2, Flatten layer and two dense layers. In the last dense layer set 10 nodes to represent the 10 digits.
9. Train the model using training samples and respective categorical labels. [use functions model.compile and model.fit]
10. Evaluate the trained model using test dataset. [Use model.evaluate]
11. Predict class of all the test samples and generate classification report.

Conclusion:

Post-lab questions:

Q.1 Draw the detailed architecture of generated CNN How many total number of trainable parameters are required for the implemented CNN.

Additional Links for reference:

- <https://machinelearningmastery.com/how-to-develop-a-convolutional-neural-network-from-scratch-for-mnist-handwritten-digit-classification/>
- <https://www.kaggle.com/code/faressayah/cifar-10-images-classification-using-cnns-88>
- <https://www.analyticsvidhya.com/blog/2020/02/learn-image-classification-cnn-convolutional-neural-networks-3-datasets/>

Note

- Write down the Expt Number, Title, PRN number, Name, and Date of performance in the first cell of the python code using markdown.
- Write down the "Result and Conclusion" in the cell next to the python code using markdown.
- Write down questions and answers in the last cell of the python code notebook using markdown.